

DP024

Physics 2

Semester 2

Session 2023/2024

2 hours



UNIT FIZIK
KOLEJ MATRIKULASI KELANTAN

PRA-PEPERIKSAAN SEMESTER PROGRAM MATRIKULASI 3
MATRICULATION PROGRAMME PRE-EXAMINATION

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU
NO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO

NAME: _____ PRAC: _____ MATRIC NO: _____	NO	Full Marks	Mark(s)
	1		
	2		
	3		
	4		
	5		
	6		
	Total		

Kertas soalan ini mengandungi 9 halaman bercetak
This question paper consists of 9 printed pages.

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ε_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|---|---|
| 1. $x = A \sin \omega t$ | 18. $\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$ |
| 2. $\omega = \frac{2\pi}{T} = 2\pi f$ | 19. $C_{eff} = C_1 + C_2 + C_3 + \dots + C_n$ |
| 3. $v = A\omega \cos \omega t$ | 20. $\tau = RC$ |
| 4. $a = -A\omega^2 \sin \omega t$ | 21. $Q = Q_0 e^{\frac{-t}{RC}}$ |
| 5. $k = \frac{2\pi}{\lambda}$ | 22. $Q = Q_0 \left(1 - e^{\frac{-t}{RC}} \right)$ |
| 6. $v = f\lambda$ | 23. $B = \frac{\mu_o I}{2\pi r}$ |
| 7. $y(x, t) = A \sin(\omega t \pm kx)$ | 24. $B = \frac{\mu_o NI}{L}$ |
| 8. $v = \frac{dy}{dt}$ | 25. $B = \frac{\mu_o I}{2r}$ |
| 9. $y = 2A \cos kx \sin \omega t$ | 26. $\phi = BA \cos \theta$ |
| 10. $v = \sqrt{\frac{T}{\mu}}$ | 27. $\Phi = N\phi$ |
| 11. $I = \frac{dQ}{dt}$ | 28. $\varepsilon = -\frac{d\phi}{dt}$ |
| 12. $Q = ne$ | 29. $\varepsilon = Blv \sin \theta$ |
| 13. $\rho = \frac{RA}{l}$ | 30. $\varepsilon = -NA \frac{dB}{dt}$ |
| 14. $R_{eff} = R_1 + R_2 + R_3 + \dots + R_n$ | 31. $\varepsilon = -NB \frac{dA}{dt}$ |
| 15. $\frac{1}{R_{eff}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \dots + \frac{1}{R_n}$ | 32. $R = 2f$ |
| 16. $V = IR$ | |
| 17. $C = \frac{Q}{V}$ | |

$$33. \quad \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$34. \quad m = \frac{h_i}{h_o} = -\frac{v}{u}$$

$$35. \quad y_m = \frac{m\lambda D}{d}$$

$$36. \quad y_m = \frac{(m + \frac{1}{2})\lambda D}{d}$$

$$37. \quad \Delta y = \frac{\lambda D}{d}$$

Answer **all** questions.

1. (a) The progressive wave with an amplitude 2.0 cm, a frequency 8 Hz and wavelength of 12 cm is moving to the left.

- (i) Calculate the period.
(ii) Write the progressive wave equation.

[5 marks]

- (b) A progressive wave can be represented by the equation

$$y(x,t) = 40 \sin (8\pi t - 6x)$$

Where y and x are in centimeter and t in seconds. Determine

- (i) direction of wave propagation,
(ii) amplitude,
(iii) frequency,
(iv) wavelength,
(v) wave propagation velocity and
(vi) particle vibration velocity at $x = 3$ cm and $t = 3$ s.

[11 marks]

2. (a) Charge passing through a conductor of cross section area 0.3 m^2 is given by

$$Q = 4t^2 + t + 5$$

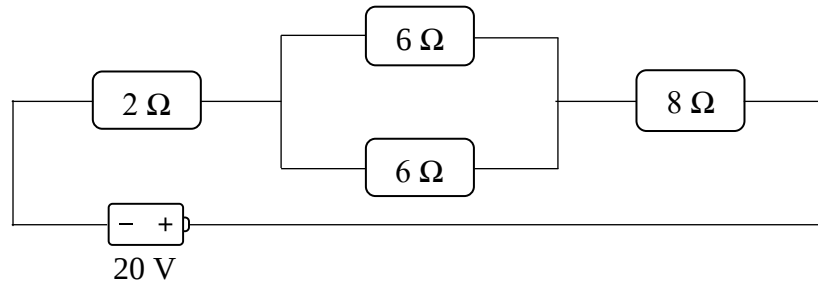
Where Q is charge in coulomb and t is time in seconds. Calculate its current at time $t = 15$ s.

[3 marks]

- (b) (i) Calculate the resistance per unit length of a copper, which has a diameter of 0.6 mm. (Given resistivity of copper is $1.68 \times 10^{-8} \Omega \text{ m}$)
(ii) If the potential difference of 10 V is maintained across a 150 cm length of the copper wire for (b)(i), what is the current in the wire?

[4 marks]

(c) (i)

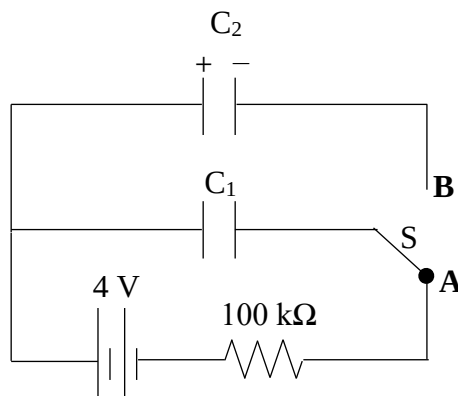
**FIGURE 1**

Find the equivalent resistance for the circuit as shown in **FIGURE 1**.

- (ii) If potential difference of 20 V is applied across the circuit, determine potential difference across the resistor of 8 Ω.

[5 marks]

3.

**FIGURE 2**

The **FIGURE 2** shows a circuit which consists of a capacitor, $C_1 = 100 \mu\text{F}$ without charge and a capacitor, $C_2 = 10 \mu\text{F}$ with a $Q_2 = 50 \mu\text{C}$ of charge. At a time, $t = 0 \text{ s}$, the two-way switch, S is connected to **A**.

- (a) Calculate time constant, τ for the charging of C_1 .

[1 mark]

- (b) Sketch graphs of charge, Q stored in C_1 against time, t .

[2 marks]

- (c) Calculate

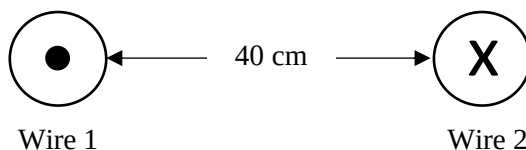
- (i) the maximum charge in C_1 .
 (ii) the amount of charge in C_1 at time $t = 20.0 \text{ s}$.

[4 marks]

- (d) When C_1 is fully charged, the switch, S is connected to **B**. Determine the potential difference across the capacitor C_2 .

[3 marks]

4.

**FIGURE 3**

- (a) Two long straight parallel wires carry 1.8 A of current in wire 1 and 2.4 A of current in wire 2, are placed 40 cm apart as shown in **FIGURE 3**.
- Determine the distance of a point from wire 1 where the resultant magnetic field is zero.
 - Determine the magnitude and direction of the magnetic field at a point halfway between the wire.

[7 marks]

- (b) A rectangular coil has sides of $2.5 \text{ cm} \times 4 \text{ cm}$ is placed in a magnetic field of 1.5 T. The angle between the vector area of a coil and the magnetic field is 30° . Calculate the magnetic flux through the coil.

[2 marks]

- (c)

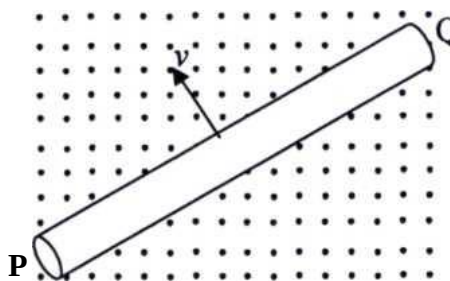
**FIGURE 4**

FIGURE 4 shows a 30 cm rod PQ moving at velocity, $v = 45 \text{ m s}^{-1}$ across a uniform magnetic field 0.8 T.

- Calculate the induced emf.
- Determine the induced current **AND** its direction if the resistance in the rod is 15Ω .

[5 marks]

5. (a) If an object is placed at 10.0 cm in front of a mirror, an upright image twice the size of the object is formed.
- (i) State the type of mirror used.
 - (ii) Determine the type of image formed.
 - (iii) Calculate the focal length of the mirror.
 - (iii) Calculate the radius of curvature of the mirror.
- [8 marks]
- (b) An image of the same height as the object is formed by a convex lens.
- (i) Find the object distance in term of the focal length.
 - (ii) Sketch a diagram to show the formation of such image.
- [6 marks]
6. (a) Two narrow slit separated 5 mm are illuminated by a light with wavelength 600 nm. The screen is placed 10 m from the slits. Determine the distance between
- (i) adjacent bright fringes on a screen.
 - (ii) the third dark fringe and fourth bright fringe.
- [9 marks]
- (b) A 550 nm light passes through two narrow slits. The interferences pattern is observed on a screen at a distance 65.0 cm from the slits. The twenty-first order bright fringes are seen at 21.2 cm apart. Calculate the
- (i) slit separation.
 - (ii) distance between two consecutive dark fringes.
- [5 marks]

END OF QUESTION PAPER
KERTAS SOALAN TAMAT